

Shadman Salif Swanan

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Research Interests

Natural Language Processing, Large Language Models, Computer Vision, OCR, Deep Learning, Semantic Drift Analysis

Technical Skills

Programming Languages: Python, C++, JavaScript, PHP

Data Analysis: MySQL, Power BI, Microsoft Excel (Advanced), Process Modeling, EDA

AI/ML Frameworks: PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Tensorflow, Hugging Face

Web & Tools: React, Vite, Node.js, Express.js, MongoDB, Git & GitHub, Tailwind CSS, Figma

Education

BRAC University <i>Bachelor of Science in Computer Science and Engineering</i>	2022 – 2026 CGPA: 3.82 / 4.00 (Merit Based Scholarship)
Notre Dame College <i>Higher Secondary Certificate (Science)</i>	2018 – 2021 GPA: 5.00 / 5.00
Ideal School and College <i>Secondary School Certificate (Science)</i>	2008 – 2018 GPA: 5.00 / 5.00 (General Scholarship/ Dhaka Board)

Research Experience

Undergraduate Thesis Research BRAC University – Conducting research on semantic drift in Bengali using diachronic word embeddings and transformer representations. Built OCR pipelines and fine-tuned Historical BanglaBERT on a 120-year corpus.	Sep 2025 – Present
Leaf Disease Research Independent – Developed TinyResNet, achieving 96.59% accuracy with distortion-aware augmentation techniques.	Apr 2026
Real-Time Fraud Detection System Independent – Built a fraud detection pipeline using LightGBM and developed a real-time transaction simulation console.	May 2026

Projects

Shikkha.com React, Vite, Node.js, Express.js, MongoDB, JWT, Stripe, Tailwind CSS – Developed a full-stack EdTech platform supporting course enrollment, online exams with automated grading, educator appointments, and e-commerce for books. Integrated JWT-based authentication, role-based access control, and Stripe Checkout.
Token-Level POS Tagging Using RNN Models Python, TensorFlow, RNN, LSTM, GRU, BiLSTM, NLP – Implemented and compared RNN, LSTM, GRU, and BiLSTM architectures for token-level part-of-speech tagging, evaluating sequence labeling performance across multiple recurrent neural network models.
Disease Prediction Using Machine Learning Python, Scikit-learn, Pandas, NumPy – Built KNN, Decision Tree, and Naive Bayes classifiers to predict diabetes from clinical features and compared model performance using standard classification metrics.
Federal Flight Agency Mission-Critical Aerospace Network Design Cisco Packet Tracer, RIPv2, Static Routing, VLSM – Designed a high-availability network for autonomous aircraft operations using VLSM and a hybrid routing strategy combining RIPv2 and static routes to ensure automatic failover and reliable communication.
Automated Sanitizer and Mask Dispensing Robot Arduino Uno, Sensors, DC Motor – Built a fully contactless hygiene system that detects users and automatically dispenses hand sanitizer and face masks using sensor-based automation and Arduino microcontroller programming.

Experience & Certifications

GCI World Program <i>The University of Tokyo (Matsuo-Iwasawa Laboratory)</i> – Participating in an international AI education program focused on machine learning and data science.	Apr 2026 – Present Remote
UX Design Decision-Making with AI & Design System Thinking with AI — GP Academy, 2026	
Conference Management Officer — Notre Dame College Model United Nations, 2019	

Co-Curricular Activities

Vice President, Photography Dept, NDNCS | Champion, BRACU Intra-Spring 2022 Football Tournament.
Runners-Up in Crusade 2026, RS Football Tournament | Completed Stage Drama Course (BRACU RS).

References

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